Hades Model Report

v0.8.4 GitHub

Hastie & Smith, et al. (2026)

MODEL CONFIGURATION

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

Mode	**Subalkaline**
Submode	**Tholeiitic**
Data suite	**./Carlingford.xlsx**
FeO/Fe^{2+}	**0.9**
Stages Run	Basic Intermediate Evolved

1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 79.76 wt.%

	Basic	Intermediate	Evolved	Total
Olivine	9.54			4.43
Clinopyroxene	24.04	24.25	18.3	23.6
Orthopyroxene		16.51		7.3
Plagioclase	58.67	41.66		45.64
K-feldspar			72.93	6.84
Magnetite	6.63	13.01	7.13	9.5
Ilmenite	1.12	4.57	1.63	2.69
FC%	37.0	56.0	27.0	

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

2. Parameter Configuration

The parameters below define the model as it was executed for this run.

	Primary	Primitive	Basic	Intermediate	Evolved
Plagioclase	1.2	1.2	0.95	0.95	1.2
Clinopyroxene	0.28	0.28	0.25	0.25	0.28
Orthopyroxene	0.29	0.29	0.29	0.29	0.29
Amphibole	0.37	0.37	0.37	0.37	0.37
K-feldspar	1.3	1.3	1.3	1.3	1.3
Biotite	1.0	1.0	1.0	1.0	1.0
Titanaugite	1.0	1.0	1.0	1.0	1.0
Nepheline	1.0	1.0	1.0	1.0	1.0
Leucite	1.0	1.0	1.0	1.0	1.0
Nosean/Hauyne	1.0	1.0	1.0	1.0	1.0
Alkali-amphibole	1.0	1.0	1.0	1.0	1.0

Table 2: Partition coefficients for the model phases.

3. Magma Evolution

Initial Compositions

	Sample	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary									
Primitive									
Basic	CS4E	50.7	1.62	15.74	13.8	4.67	9.21	2.2	2.04
Intermediate	B4	54.64	1.87	13.55	14.83	3.39	6.83	3.07	1.82
Evolved	288-8a	71.24	0.47	12.93	4.74	0.5	1.46	3.03	5.63
End	LWG2	75.18	0.21	12.91	2.19	0.2	0.74	3.01	5.56

Table 3: Major oxide compositions of the model phases (wt.%).

Computed Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary								
Primitive								
Basic	50.7	1.62	15.74	13.8	4.67	9.21	2.2	2.04
Intermediate	54.13	1.87	13.56	14.76	3.39	6.85	2.26	3.19
Evolved	70.0	0.49	12.8	4.76	0.5	1.45	2.87	7.14
End	75.06	0.21	12.05	2.2	0.23	0.66	2.97	6.61

Table 4: Major oxide compositions of the model phases (wt.%).

4. Optimisation Results

Basic > Intermediate

BASIC

Population size	50
Iterations	500/500
Cost function	1.58
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	50.7	54.64	54.13	0.93
TiO2	1.62	1.87	1.87	0.2
Al2O3	15.74	13.55	13.56	0.02
Fe2O3	13.8	14.83	14.76	0.49
MgO	4.67	3.39	3.39	0.17
CaO	9.21	6.83	6.85	0.28
Na2O	2.2	3.07	2.26	26.45
K2O	2.04	1.82	3.19	75.29

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	37.0
Olivine	9.54
Spinel	0.0
Clinopyroxene	24.04
Orthopyroxene	0.0
Plagioclase	58.67
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	6.63
Ilmenite	1.12

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	44.83	1.19	19.47	12.37	6.7	13.24	2.1	0.09

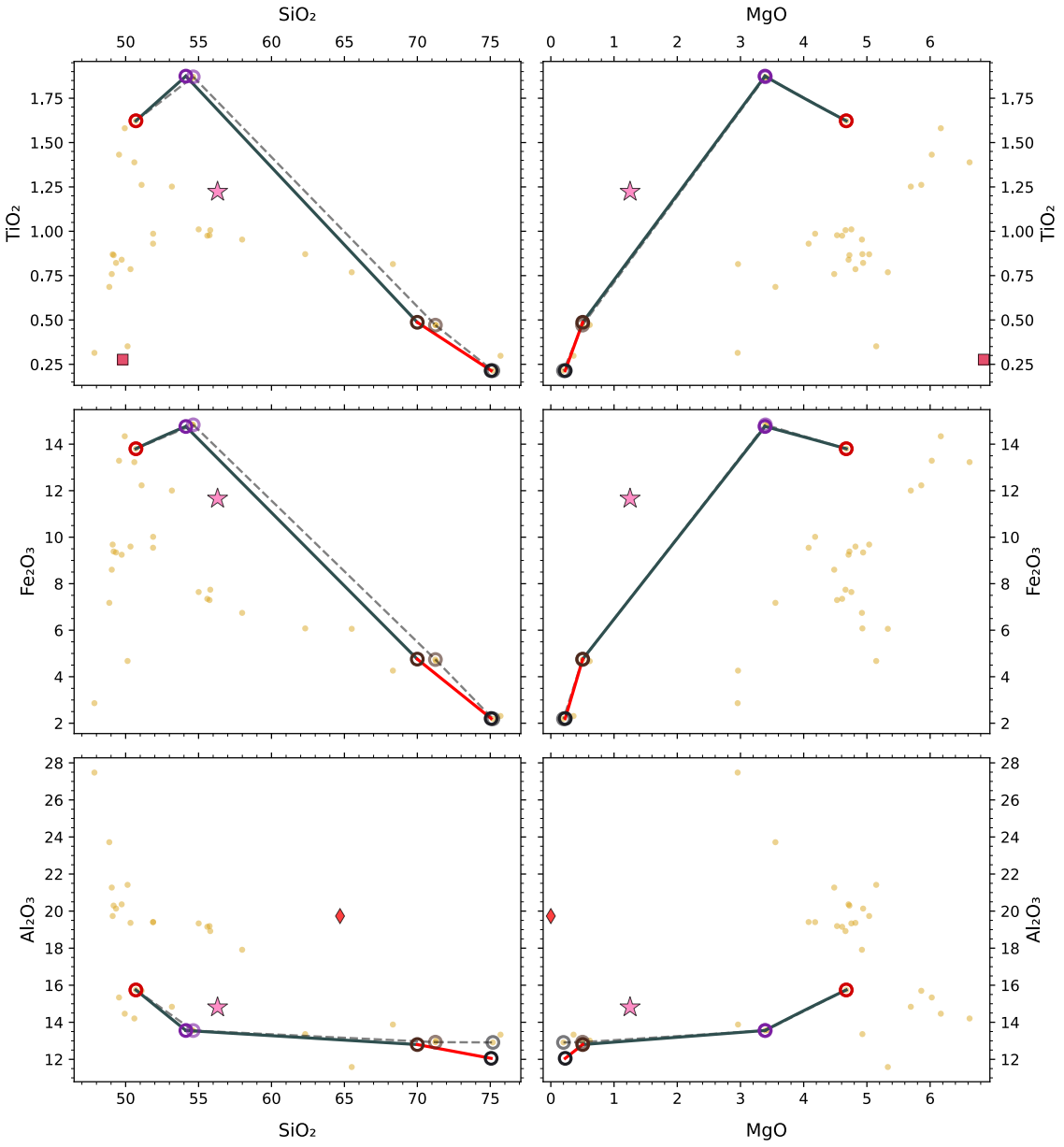
Table 7: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	36.37	0.06	0.16	29.07	33.97	0.38	0.0	0.0
spinel								
clinopyroxene	47.07	1.24	5.06	10.73	14.38	20.8	0.66	0.07
orthopyroxene								
plagioclase	51.2	0.0	31.09	0.28	0.01	13.98	3.31	0.13
amphibole								
kfeldspar								
biotite								
quartz								
magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0

Table 8: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

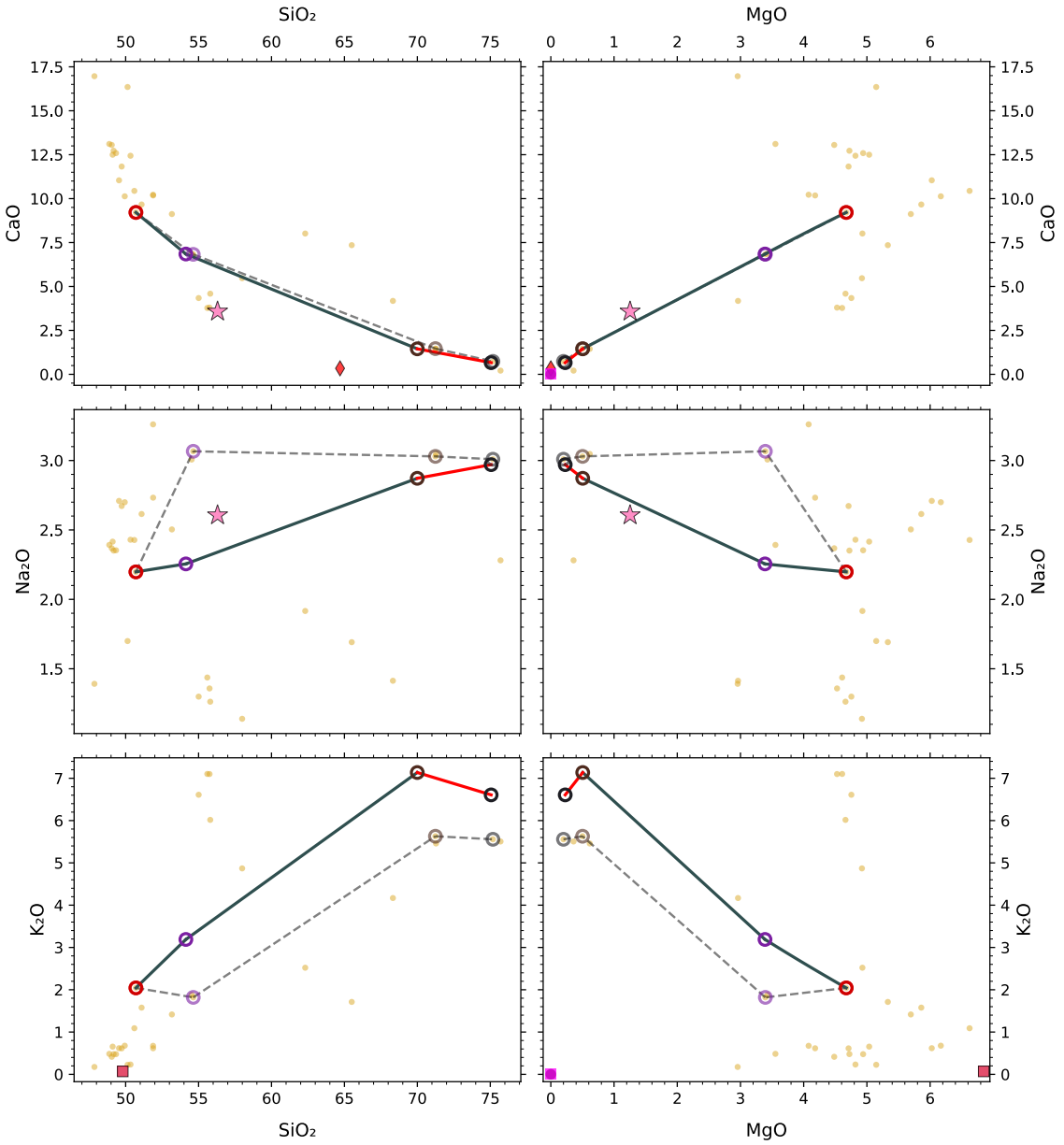


Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Basic to Intermediate transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Intermediate > Evolved

INTERMEDIATE

Population size	50
Iterations	500/500
Cost function	0.96
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	54.13	71.24	70.0	1.75
TiO2	1.87	0.47	0.49	3.4

	Initial	Expected	Computed	% Error
Al2O3	13.56	12.93	12.8	1.0
Fe2O3	14.76	4.74	4.76	0.37
MgO	3.39	0.5	0.5	0.57
CaO	6.85	1.46	1.45	0.89
Na2O	2.26	3.03	2.87	5.21
K2O	3.19	5.63	7.14	26.76

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	56.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	24.25
Orthopyroxene	16.51
Plagioclase	41.66
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	13.01
Ilmenite	4.57

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	41.66	2.96	14.15	22.62	5.65	11.09	1.77	0.09

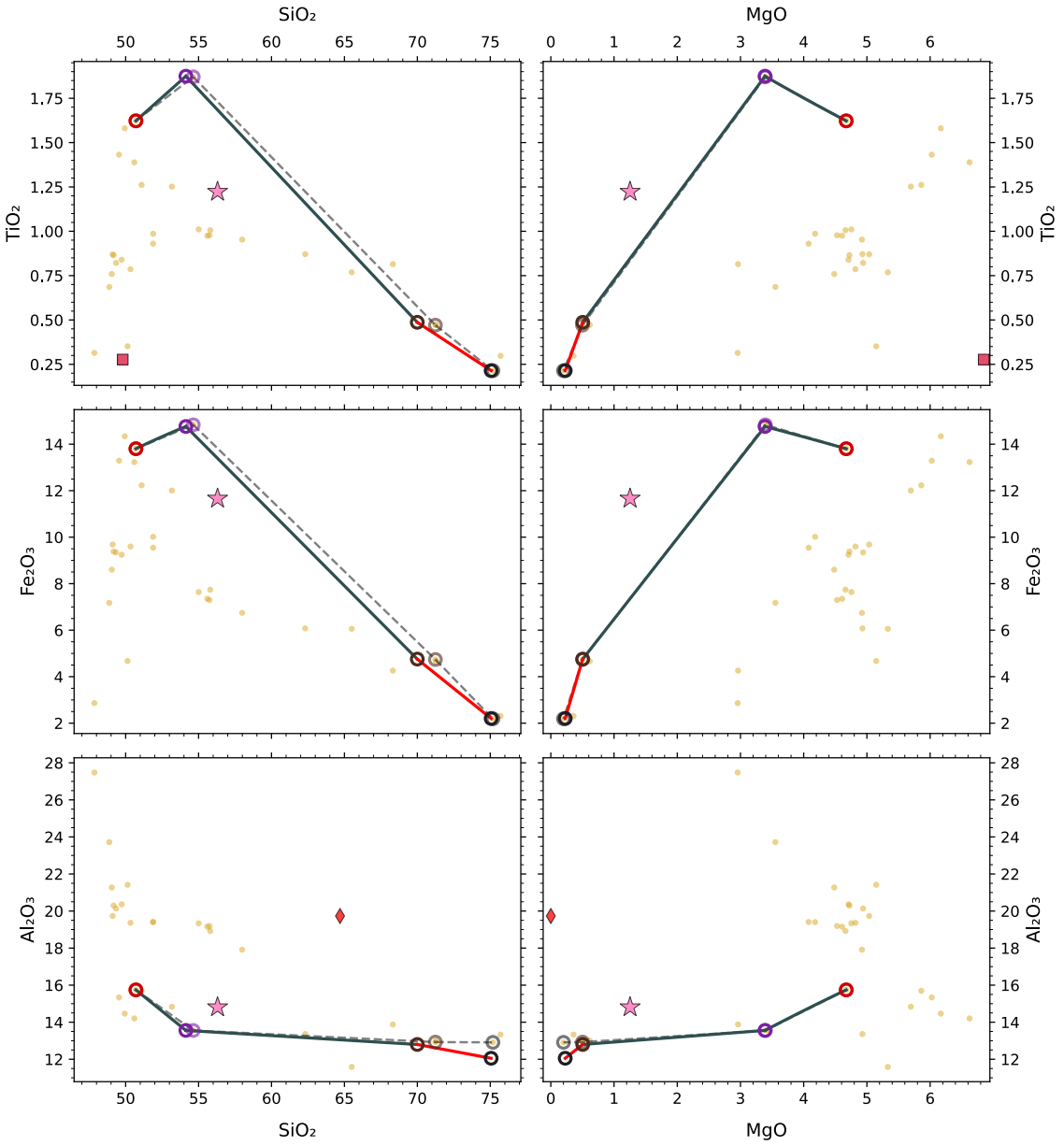
Table 11: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine								
spinel								
clinopyroxene	46.87	0.0	7.27	11.55	10.54	23.54	0.08	0.14
orthopyroxene	50.05	0.19	0.16	29.21	18.75	1.63	0.0	0.0
plagioclase	52.89	0.0	29.68	0.84	0.0	12.26	4.2	0.13
amphibole								
kfeldspar								
biotite								
quartz								
magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0

Table 12: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

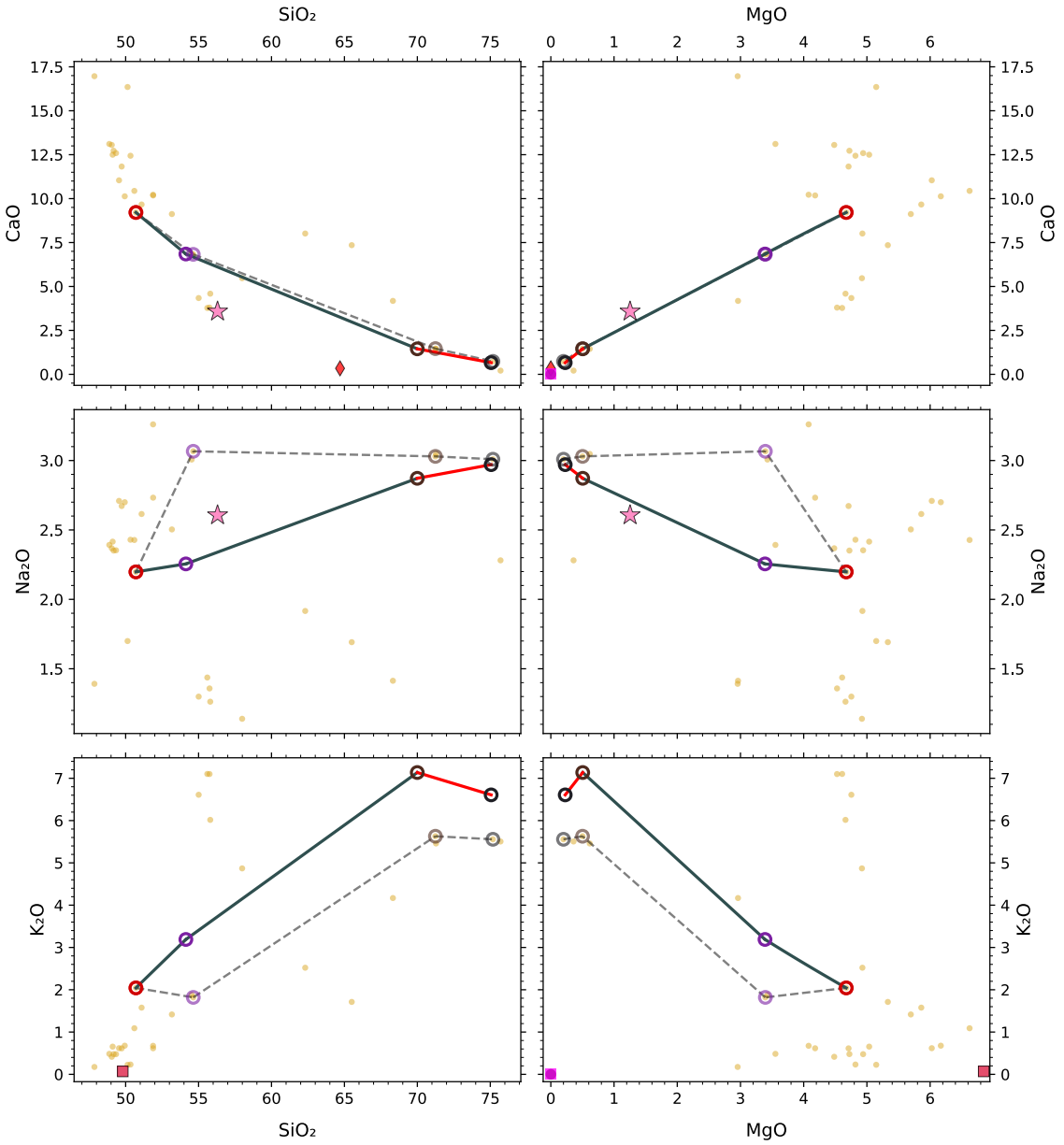


Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Intermediate to Evolved transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Evolved > End

EVOLVED

Population size	50
Iterations	500/500
Cost function	0.88
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	70.0	75.18	75.06	0.16
TiO2	0.49	0.21	0.21	0.04

	Initial	Expected	Computed	% Error
Al2O3	12.8	12.91	12.05	6.65
Fe2O3	4.76	2.19	2.2	0.62
MgO	0.5	0.2	0.23	13.72
CaO	1.45	0.74	0.66	9.89
Na2O	2.87	3.01	2.97	1.3
K2O	7.14	5.56	6.61	18.9

Table 13: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	27.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	18.3
Orthopyroxene	0.0
Plagioclase	0.0
Amphibole	0.0
K-feldspar	72.93
Biotite	0.0
Quartz	0.0
Magnetite	7.13
Ilmenite	1.63

Table 14: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	56.3	1.22	14.81	11.66	1.25	3.58	2.61	8.57

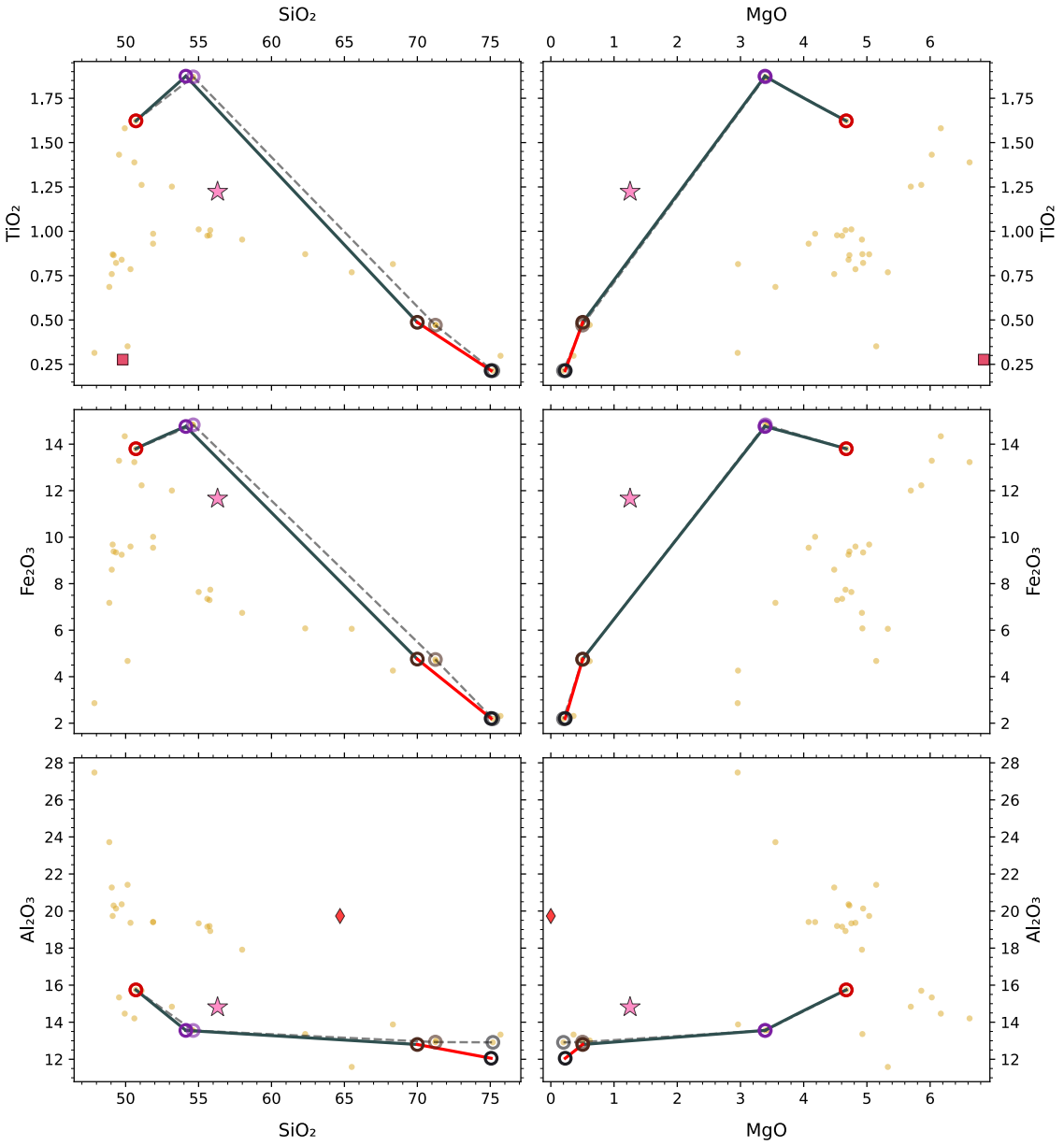
Table 15: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine								
spinel								
clinopyroxene	49.8	0.28	2.3	21.91	6.85	18.2	0.6	0.07
orthopyroxene								
plagioclase								
amphibole								
kfeldspar	64.7	0.0	19.73	0.08	0.0	0.34	3.42	11.73
biotite								
quartz								
magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0

Table 16: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

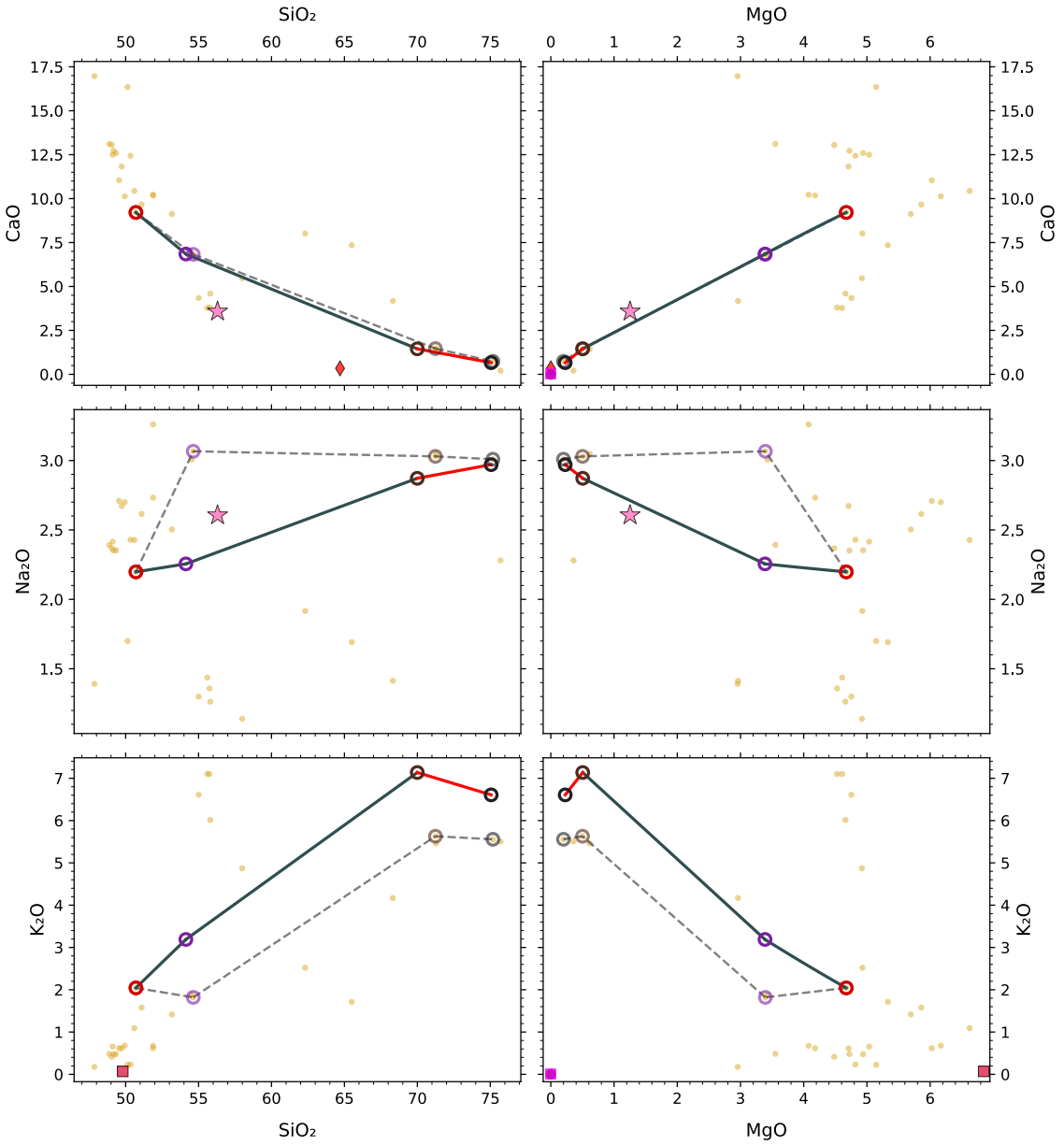


Figure 3: Bivariate plots of the data suite with the data-based and modelled LLODs. The Evolved to End transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

5. Figures

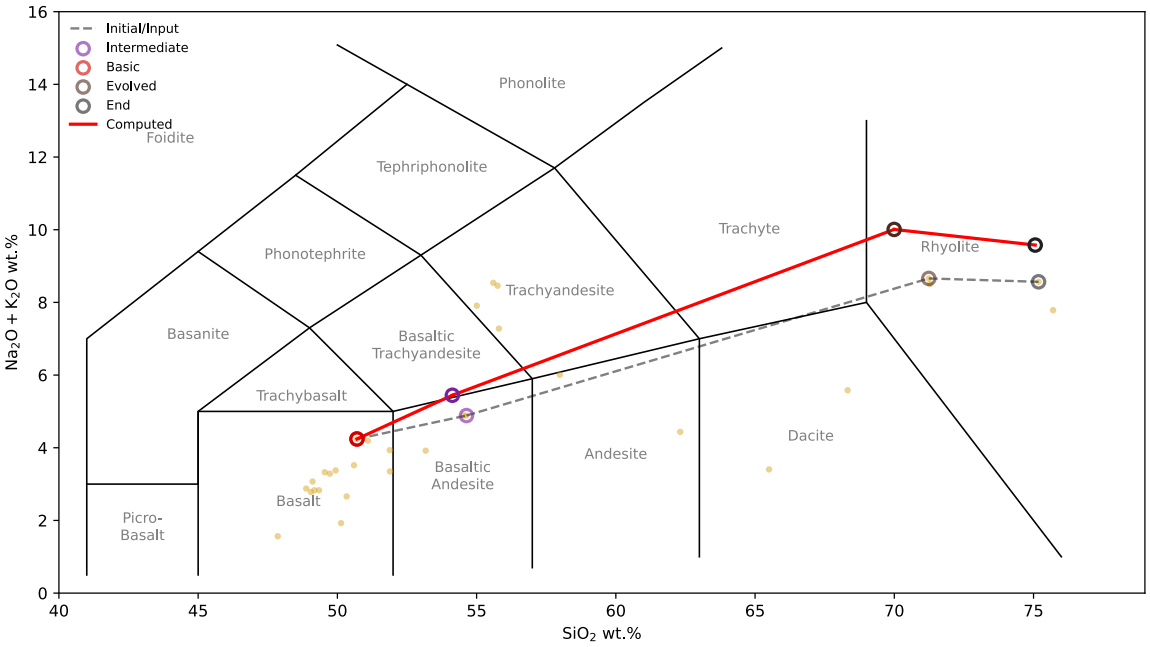


Figure 4: Total Alkali-Silica (TAS) diagram of the model phases.

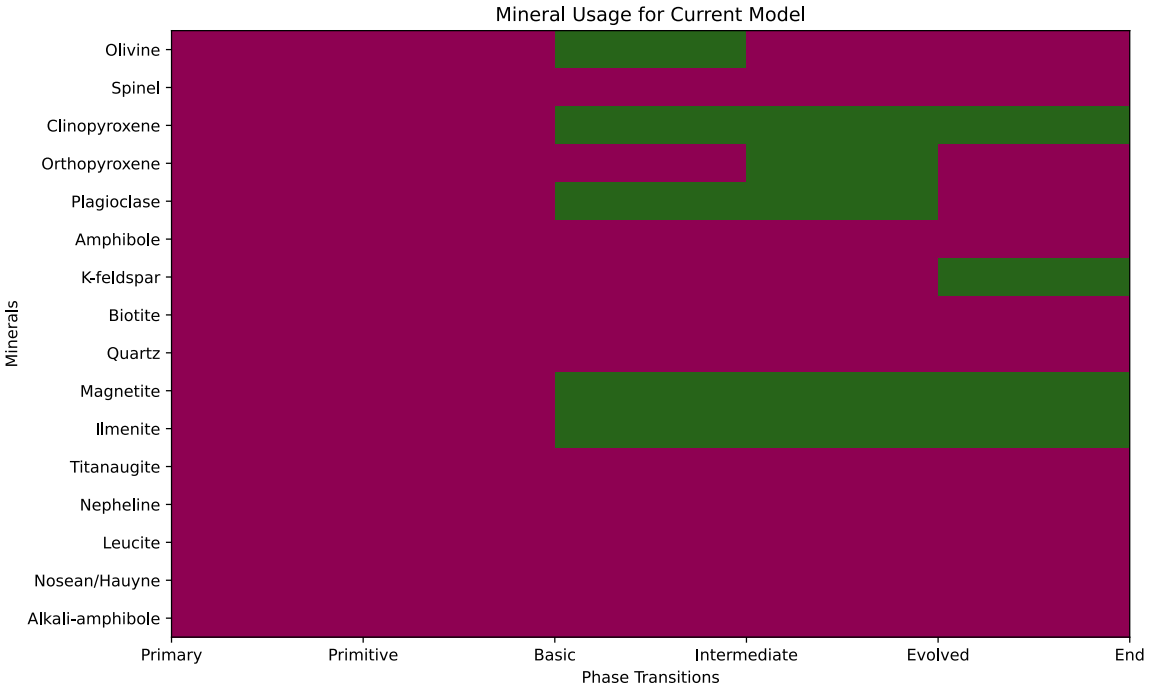


Figure 5: Mineral usage across the model phases.

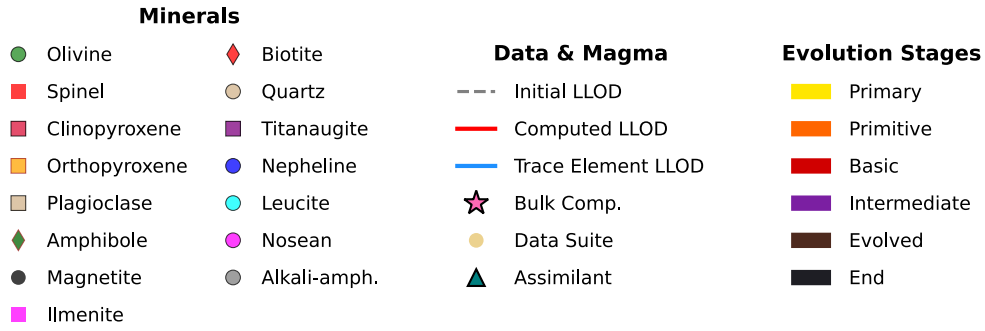


Figure 6: Legend for mineral symbols used in the figures.